ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

Internet of Things (IoT)

Lecture 3:
Smart Objects & IoT Devices

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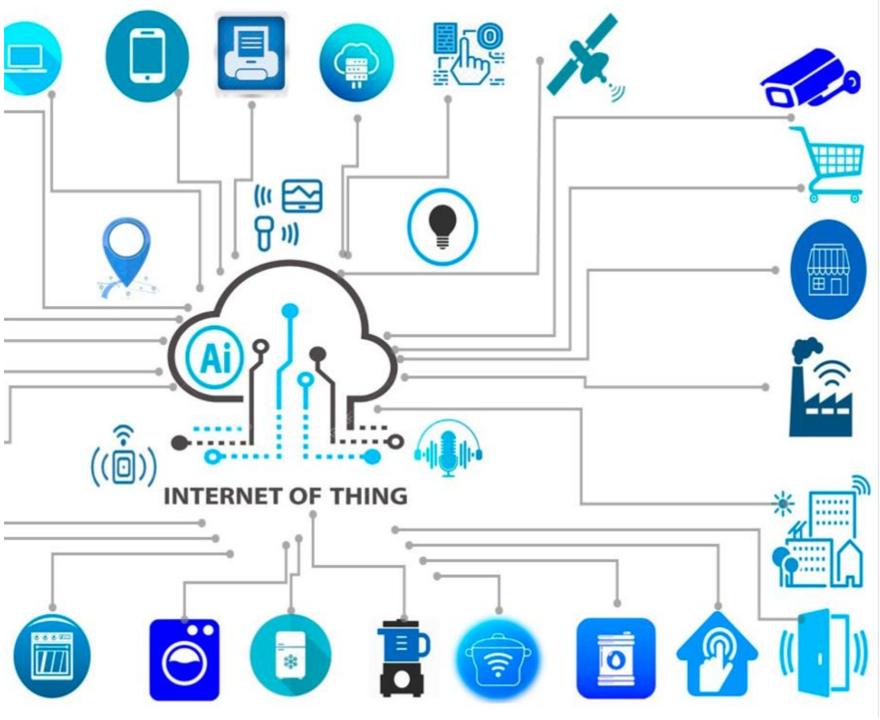
IoT Course Topics

1. Introduction to IoT
2. IoT Architecture & Network Components
3. Smart Objects & IoT Devices
4. Communication Technologies in IoT
5. IoT Protocol Stacks
6. IoT Application Protocols & Data Formats
7. IoT Security Principles
8. Cloud & Fog Computing in IoT
9. Data Analytics in IoT
10. IoT Platforms & Middleware
11. Designing IoT Solutions



Smart Objects & IoT Devices

Agenda

- IoT Architecture
- Traditional Data Flow in IoT
- Data Processing Locations
- IoT Ecosystem
- IoT Framework
- Things
- Sensors
- Actuators

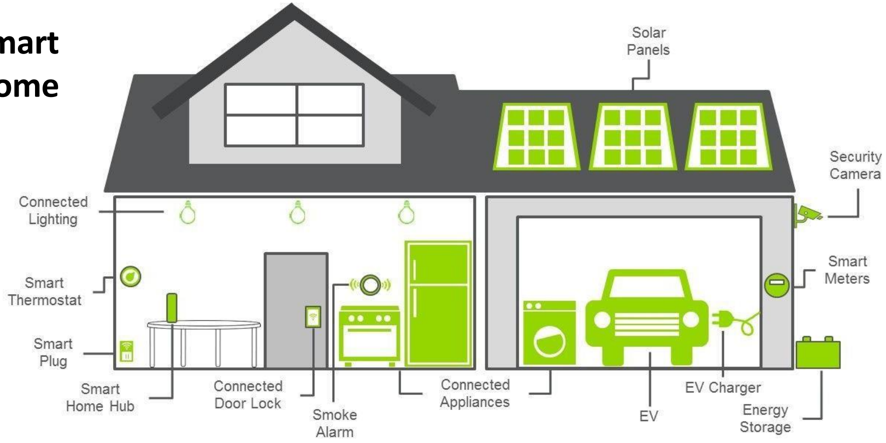
What is Architectural Plan?

- In present days, networks run the modern business

- So, it should never be built without careful planning

  - Architecture is <u>how you design</u> your application or solution.

Smart Home

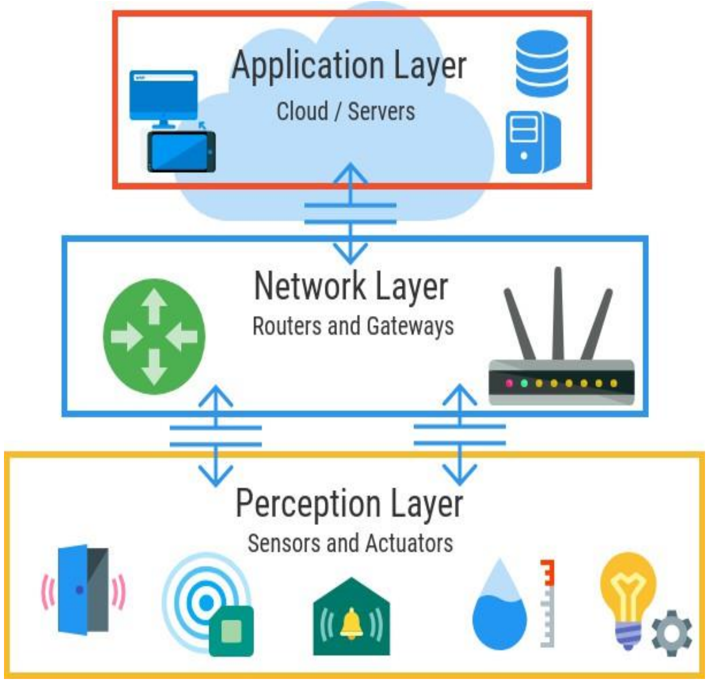


- Essence of IoT architecture:
  - how the data is
    - transported,
    - collected,
    - analyzed, and
    - ultimately acted upon.

Driving forces:

- Scale
- Security
- Constrained devices
- Massive data
- Data analysis
- Support to legacy devices

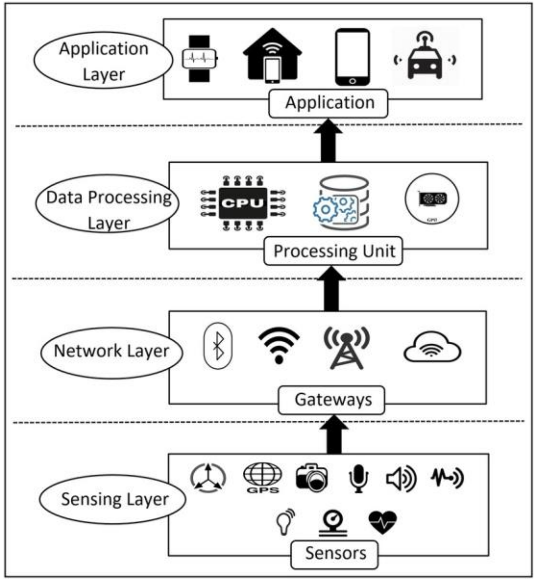
Basic 3-Layer architecture



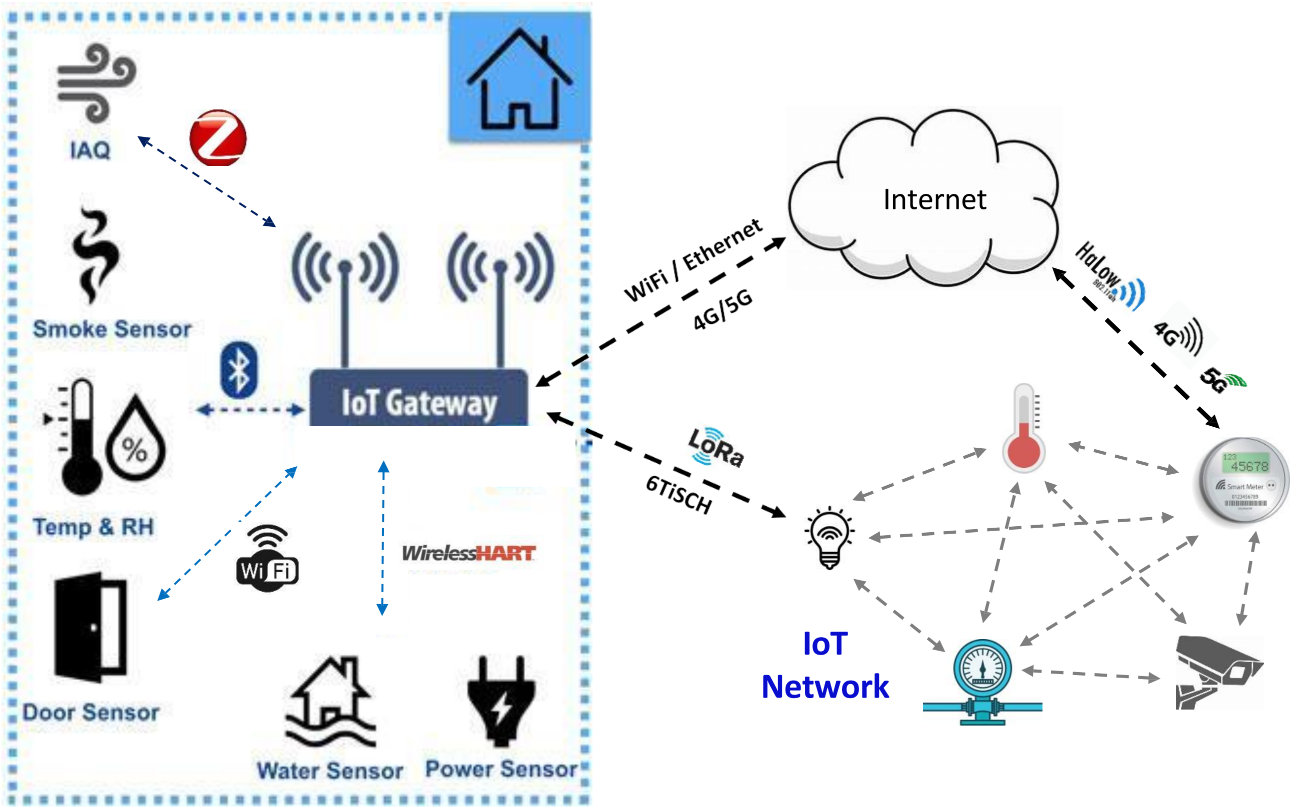
- **Perception layer** is the physical layer, which has sensors for sensing and gathering information about the environment.
- **Network layer** is responsible for connecting to other smart things, network devices, and servers. Its features are also used for transmitting and processing sensor data.
- **Application layer** is responsible for delivering application specific services to the user.
  - For example, smart homes, smart cities, smart health, etc.

Commonly used 4-Layer architecture

- **Sensing layer:** This layer includes sensors and actuators that are placed in the environment to gather information
- **Network layer:** It includes protocols and technologies that enable devices to connect and communicate with each other and with the wider internet.
- **Data processing Layer:** It includes a variety of technologies and tools, such as data management systems, analytics platforms, and machine learning algorithms, to extract meaningful insights from the data and make decisions based on that data.
- **Application Layer:** It includes various software and applications such as mobile apps, web portals, and other user interfaces that are designed to interact with the underlying IoT infrastructure.

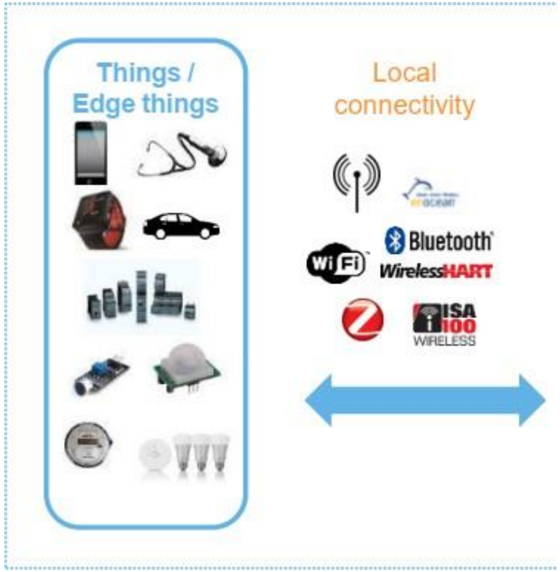


Smart Home IoT Application

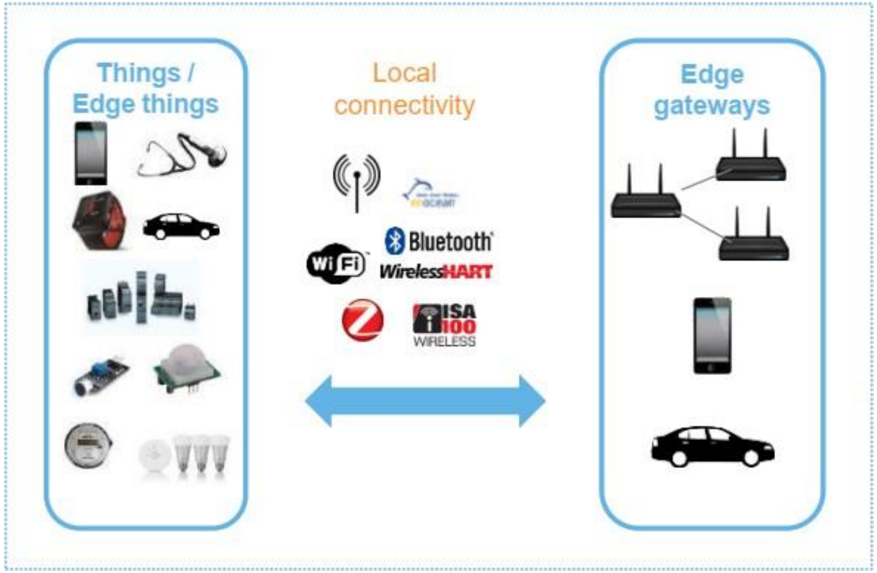


Traditional Data Flow in IoT

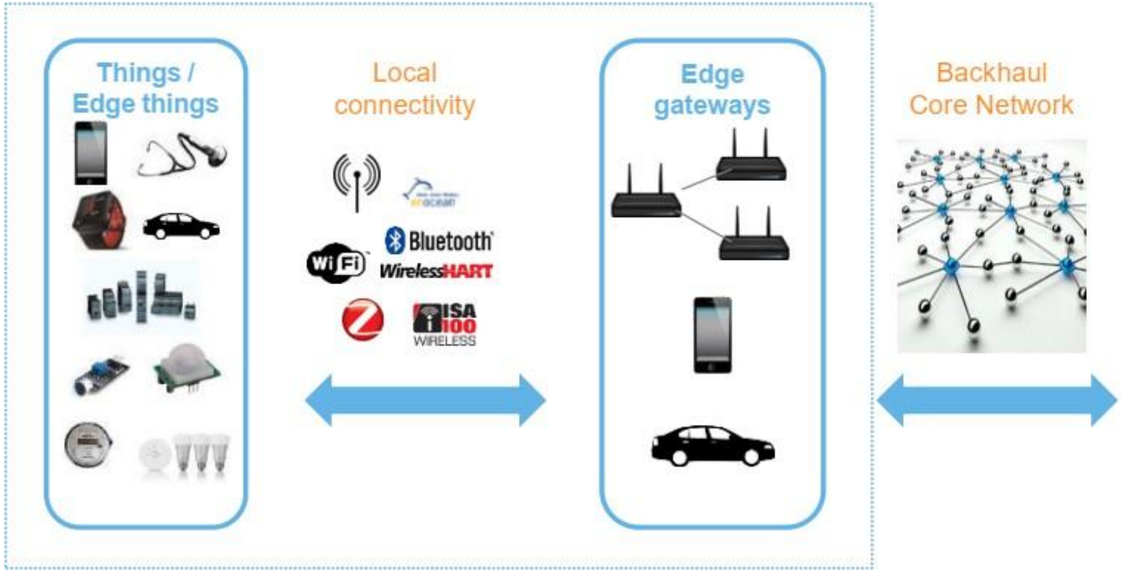
Traditional Data Flow in IoT



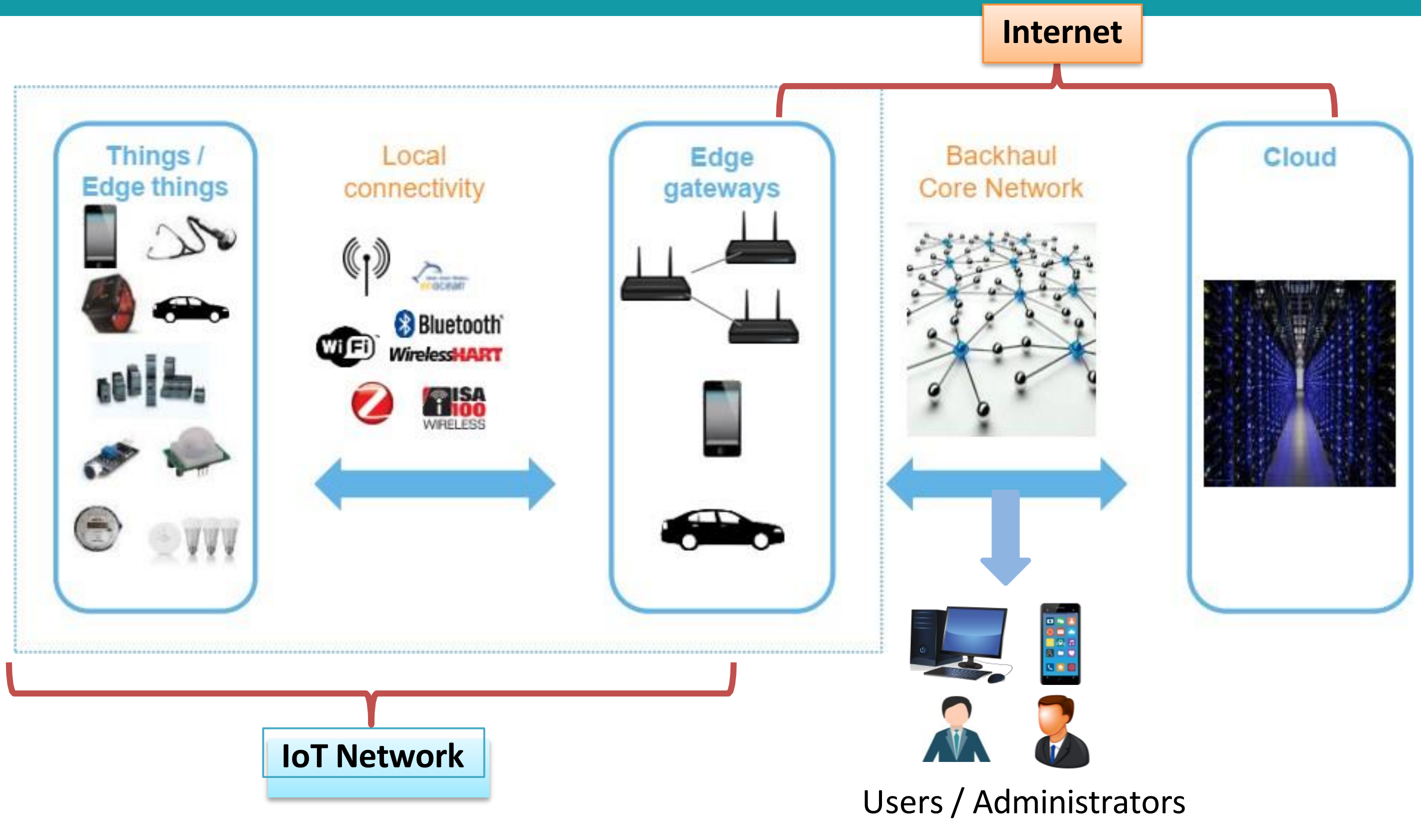
Traditional Data Flow in IoT



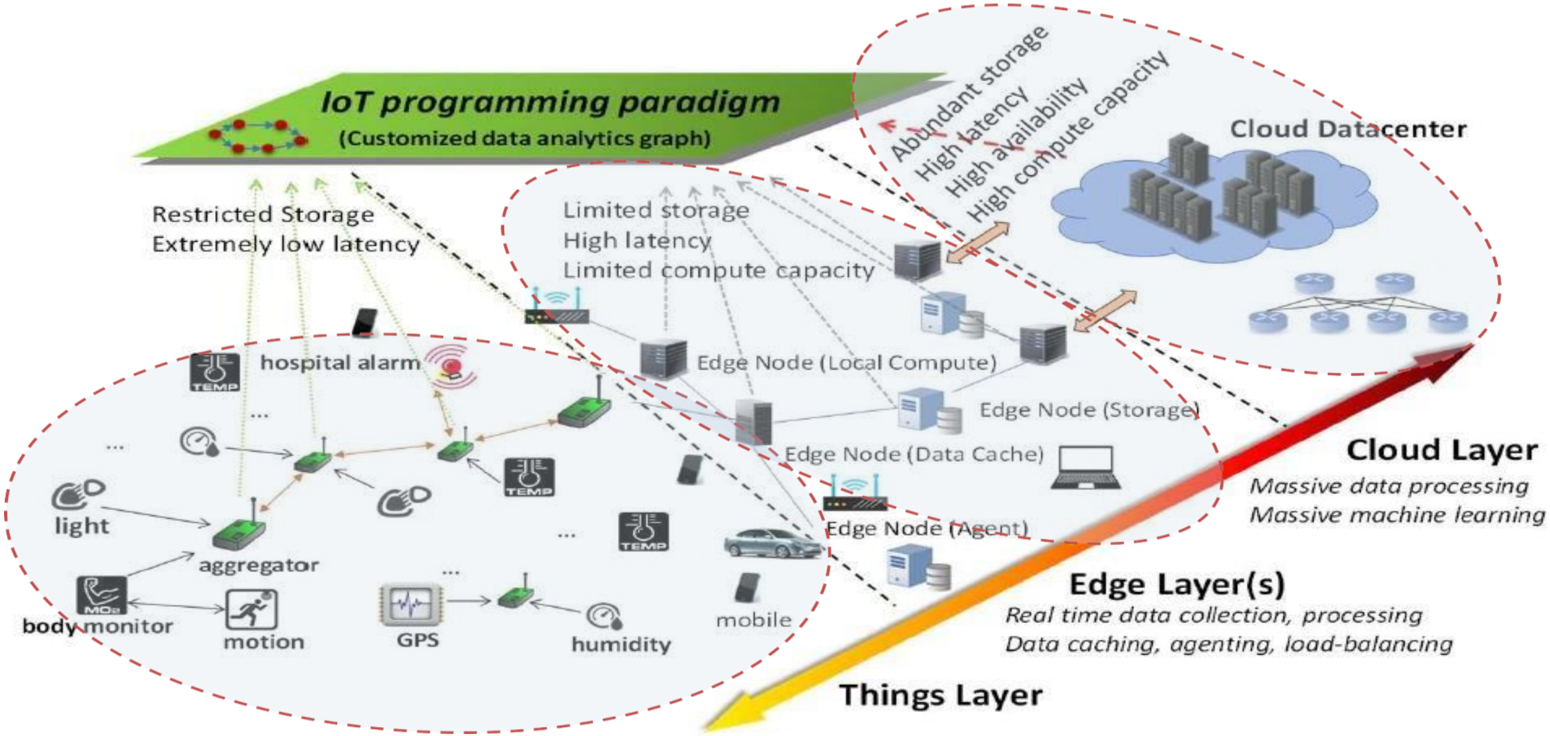
Traditional Data Flow in IoT



Traditional Data Flow in IoT



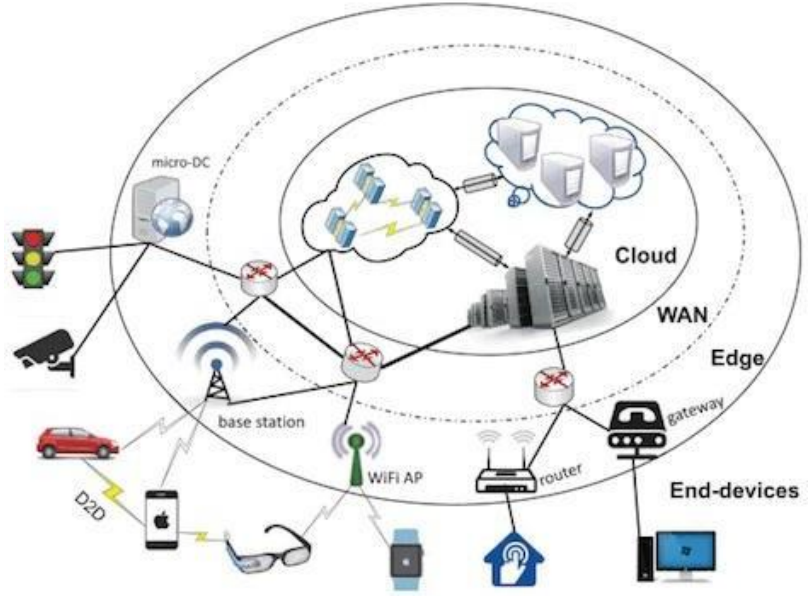
Data Processing Locations



Data Processing at the Edge

Edge Computing

- Host computation tasks close to the data sources and end users.



AI:

- Extracts insights that leads to better decisions & strategic move

Edge Intelligence:

- Apply AI on data at network Edge

What is needed to know?

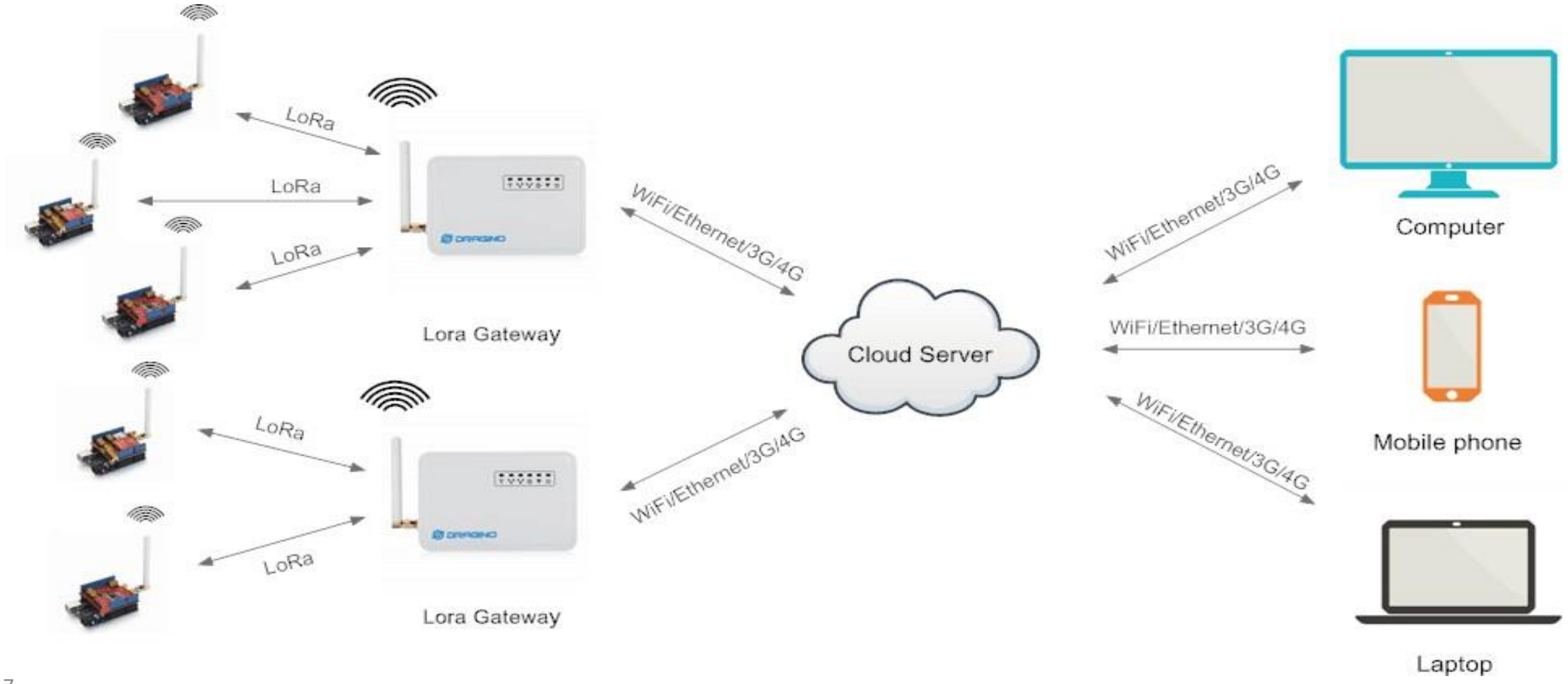
- **IoT Components**
  – Application developer needs to know what are the required components for designing an IoT application

- **IoT Architecture**
  _ For any IoT application it is required to know the physical arrangement of IoT sensors, actuators, gateway, edge devices and Internet connectivity

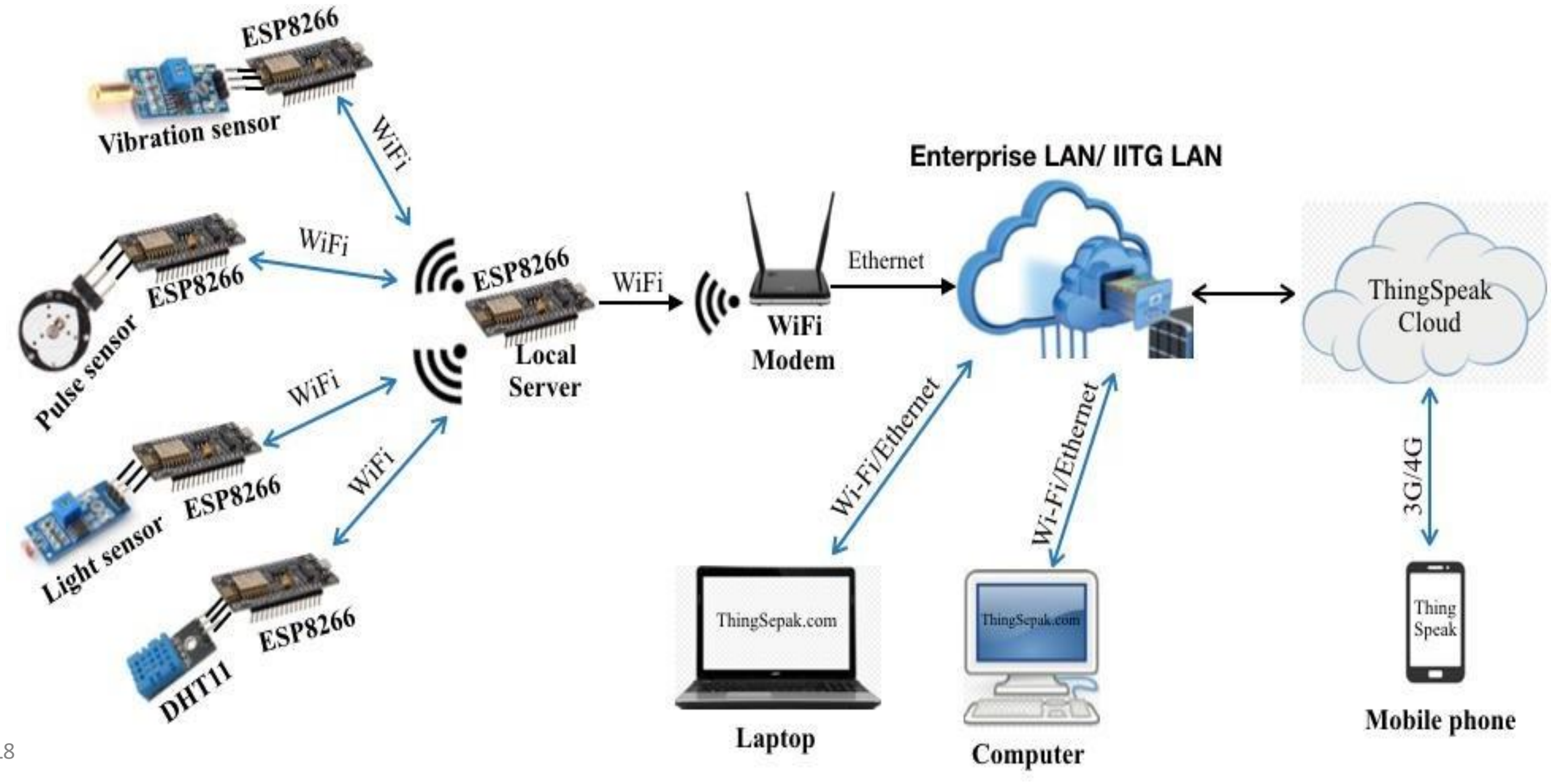
- **IoT Network Access Technology**
  – Types of IoT Network Access Technology or Connectivity Technology used to connect the sensors/actuators with each other, and with the IoT gateway or Internet

- **IoT Data Processing / Analysis**
  – Sensed data needs to be stored and analyzed to take business decision or to get new business insight. Where it can be done and How?

DIY: Long-range Monitoring & Control



DIY: Smart Home Monitoring



IoT Ecosystem

It encompasses **all the components** needed to collect, process, and analyze data from IoT devices, enabling smart applications and services.

✓ IoT Core
- Sensors & Actuators, microcontrollers, internet connectivity, service platform including security

✓ IoT Gateway
- It ensure bidirectional communication between IoT networks and other networks

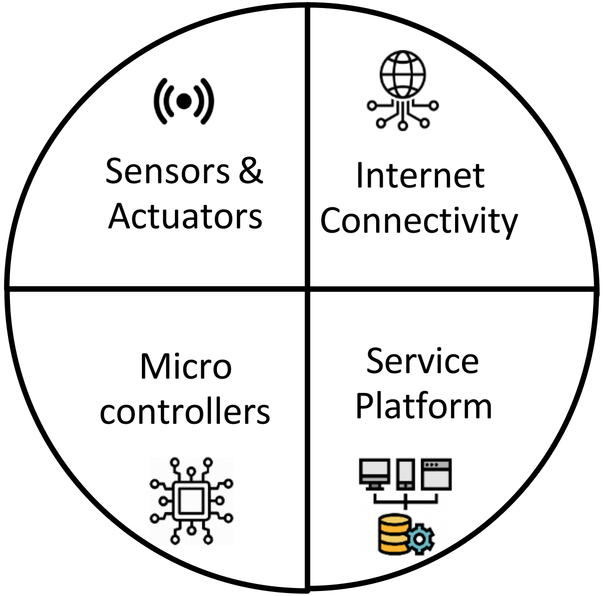
✓ Cloud
- Accepts, accumulates, maintains, stores, and process data

✓ Analytics
- It indulges in conversion and analysis of data which results in recommendations and future decision making

✓ User Interface / Visualization
- Design sleek, visually appealing, interactive, and ease-of-use graphical user interface (GUI)

✓ IoT Architectures
- Graphical structure of the designed IoT-based solutions and products
  - 3-layer architecture, IoT-WF, oneM2M, etc.

Core Components of IoT

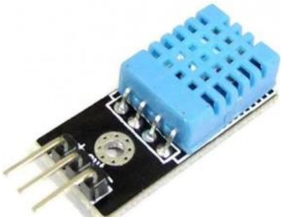


- Sensors - to gather data and events
- Actuators – responsible for moving and controlling a mechanism or system
- Microcontrollers - automatically controls sensors and actuators; makes them smart
- Internet connectivity – responsible for sharing information and control command
- Service Platform – ability to deploy and manage the IoT devices and applications including data management, data analytics and security

“Things” in IoT – Sensors

MQ135 - Air Quality Gas Sensor

Sound Detection Sensor



DHT11 - Temperature and Humidity Sensor

PIR Motion Detector Sensor

Pulse Sensor

LDR Light Sensor

Ultrasonic Distance Sensor

IR Sensor

“Things” in IoT – Actuators

4 Channel 5V Relay

Servo Motor

DC Motor

Solenoid valve

Linear Actuators

LED

LCD Diplay

Access Technologies in IoT

Communication Criteria

- Range
- Frequency Bands
- Power Consumption
- Topology
- Constrained Devices
- Constrained-Node Networks

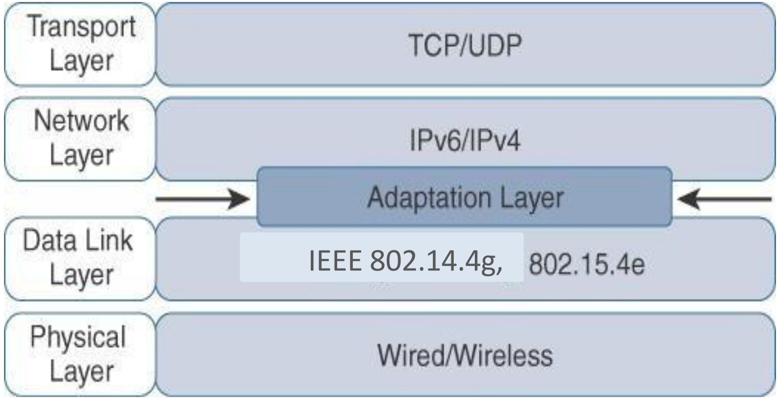
IoT Access Technologies

LoRaWAN™
ZigBee®
Sub-1GHz
6LoWPAN
Bluetooth®
sigfox
NB-IoT™
NFC
Z-WAVE®
Wi Fi
THREAD
nwave
5G

Use of Internet Infrastructure

Key Advantages of IP

- ➢ Open and standard-based
- ➢ Versatile
- ➢ Ubiquitous
- ➢ Scalable
- ➢ Manageable
- ➢ Highly secure
- ➢ Stable and resilient

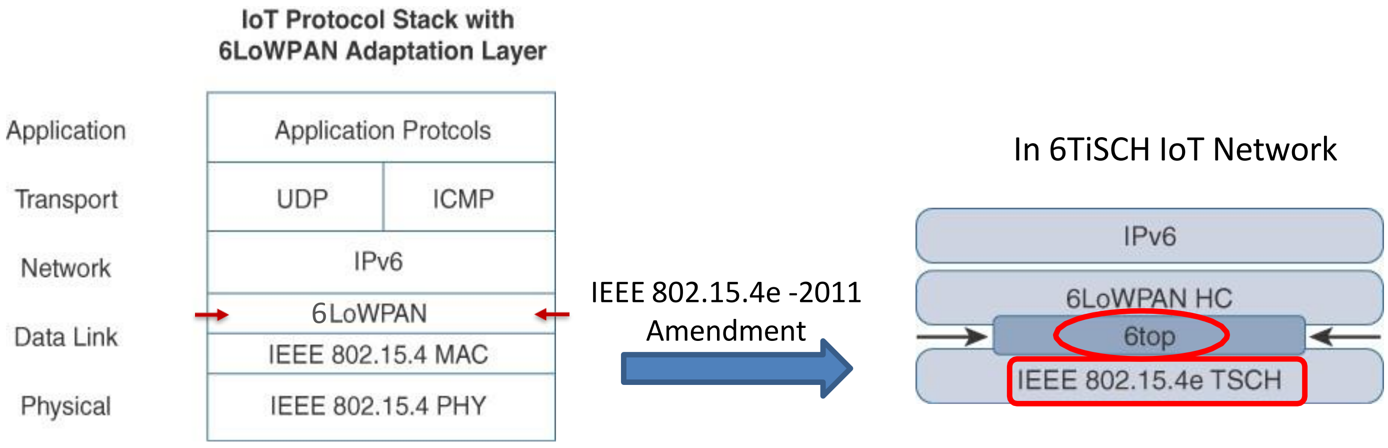


- IPv6 packets require a minimum MTU/PDU size of 1280 bytes.
- The maximum size of a MAC layer frame in IEEE 802.15.4 is 127 bytes.
  - It gives just 102 bytes for an IPv6 packet !!

Need of packet/frame size optimization due to

- ➢ Constrained Nodes
- ➢ Constrained Networks

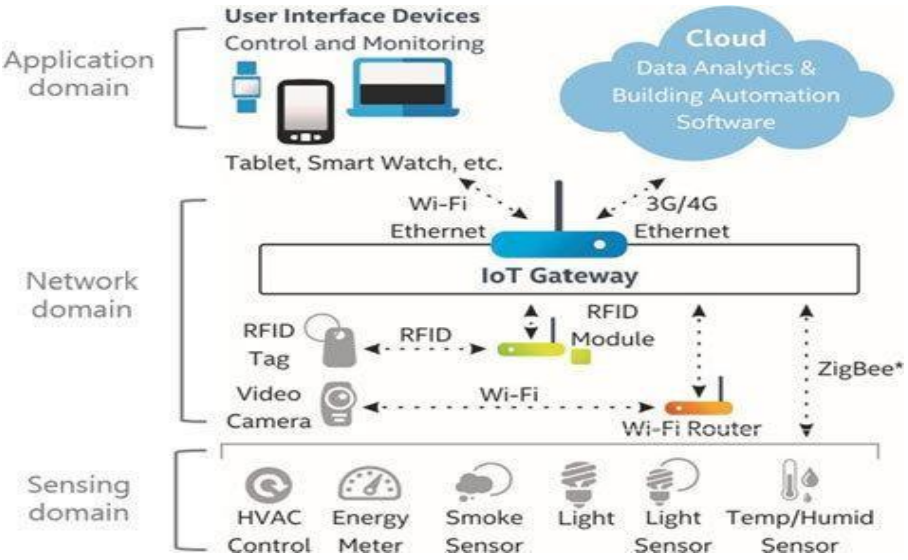
Modification in Stack



- WPAN: Wireless Personal Area Networks
- IEEE 802.15.4: Low-Rate WPAN
- 6LoWPAN: IPv6 over Low-Power WPAN
- TSCH: Time Synchronized Channel Hopping
- 6TiSCH: IPv6 over the TSCH mode of IEEE 802.15.4e
- 6top: 6TiSCH Operation Sublayer

IoT Gateway

- It is a physical device or software program that serves as the connection point between the two different types of networks



Gateway

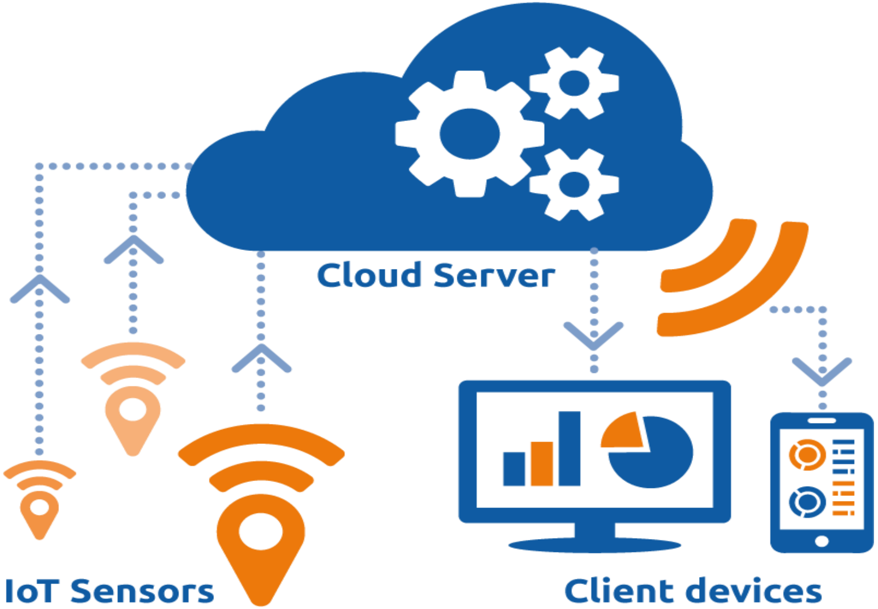
- Provides bidirectional communication
  - Between IoT protocols and other networks
    - e.g. Zigbee <--> Ethernet
- Sometimes programmed to execute some processing operations
  - Edge computing
- It is necessary to maintain security to a certain extent
  - Can shield the entire IoT systems from any cyberattack

Use of Cloud

- IoT generates vast amount of Big Data; this in turn puts a huge strain on Internet Infrastructure.

- Cloud can facilitate to
  - Provide different services
  - Store huge amount of data
  - Process the data efficiently

- **Benefits of Cloud Platform in IoT**
  - Network Scalability
  - Data Mobility
  - Time to market
  - Security
  - Cost-effectiveness

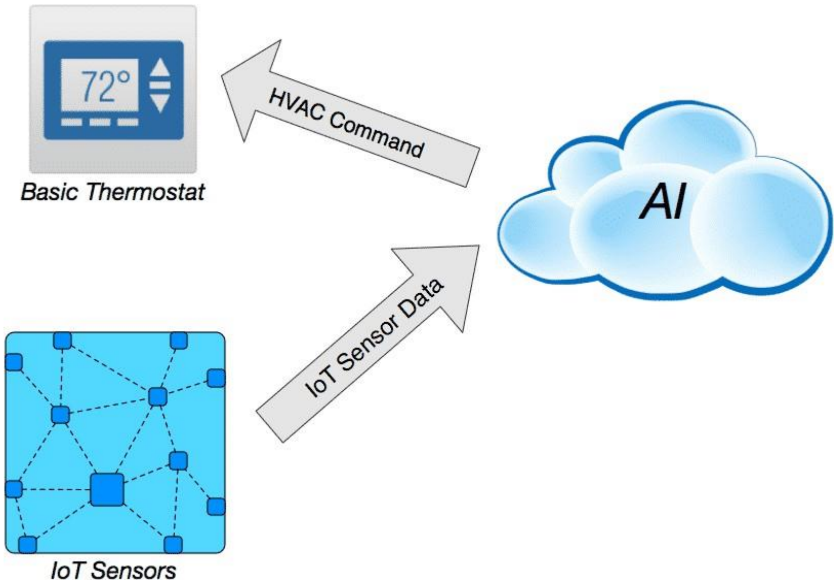


AI for IoT

- AI focuses on putting human intelligence in machine
- It gives the ability to a machine/program to think and learn by itself

Use of AI in IoT:

- Smart Home
  - Automated HVAC control
- Industrial IoT
  - Predictive maintenance
  - Optimized supply chain
- Farming
  - Smart farming
  - Interruption warning
- Self-driving Car
  - Mimic human driving on road
- Health
  - Auto-diagnosing any disease
  - Assistive healthcare

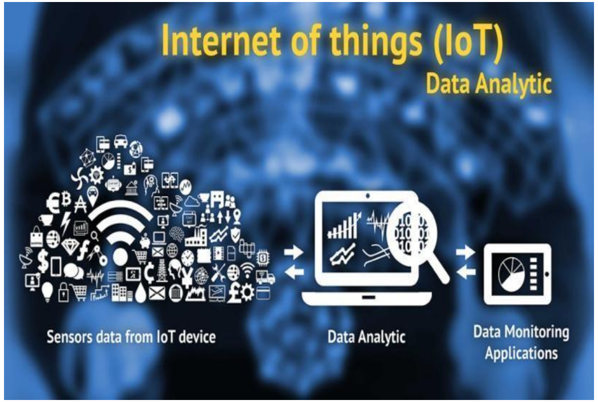


Data Analytics in IoT

❖ The business value of IoT is not just in the ability to connect devices, but it comes from understanding the data these devices create.

"Data Analytics + IoT => Smart Business Solutions"

➢ IoT analytics is the application of data analysis tools and procedures to realize value from the huge volumes of data generated by connected IoT devices



Challenges:

✓ Huge Volume

✓ Real-time data flow

✓ Variety of data types
  - e.g. XML, video, SMS

✓ Unstructured data

✓ Variable data model and meaning / value

Securing IoT

- Both the IoT manufacturers and their customers didn't care about the security !

Unauthorized access to IoT devices

Source:https://www.theguardian.com/technology/2016/oct/26/ddos-attack-dyn-mirai-botnet

Major cyber attack disrupts internet service across Europe and US;
October 26, 2016

Unauthorized access to IoT network

Source:http://metropolitan.fi/entry/ddos-attack-halts-heating-in-finland-amidst-winter

DDoS attack halts heating in Finland amidst winter;
November 7, 2016

User Interface

- Information made available to the end-users
- Users can actively check and act in for their IOT system

Important Characteristics:

- ✓ Sleek design
- ✓ Visually appealing
- ✓ Interactive UI
- ✓ Ease-of-use
- ✓ Handy

IoT Framework

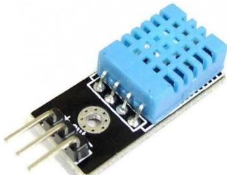
- Framework provides a development environment.
  - It provides appropriate infrastructure to design and implement the architecture

- IoT framework comprises of large number of components
  - sensors, sensor systems, gateways, mobile app, embedded controller, data management platform, analytical platform, and so on.
  - support interoperability among all devices, provides secure connectivity, reliability in data transfer, interface to 3rd party application to built on it, and so on.

Few IoT Framework	Few IoT Framework
RTI (Real-Time Innovations) Connext DDS	Cisco Ultra IoT
Salesforce IoT cloud	Microsoft Azure IoT
Eclipse IoT	PTC ThingWorx
GE (General Electronic) Predix	Amazon AWS IoT
IBM Watson IoT	Kaa

The “Things”

- **Sensors & Actuators are the fundamental building blocks of IoT**
  - Sensor senses
  - Actuator acts

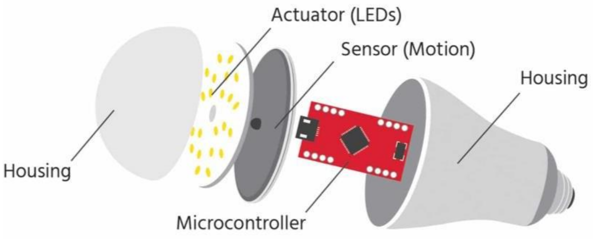


Sensor

- **Smart objects are any physical objects that contain**
  - Embedded technology
  - Microcontroller unit, memory storage, power supply, communication ports, input and output, timer or counter
  - Sensors and/or actuators

Actuator

- Smart objects are to sense and/or interact with their environment in a meaningful way
  - being interconnected, and
  - enabling communication among themselves or with external agent.



Smart Object

Sensors

- It **measures some physical quantity** and converts that measurement into analog/digital form

- There are a number of ways to group and cluster sensors into different categories

- Based on external energy requirement
  - Active / Passive

- Based on placement location
  - Invasive / Non-invasive

- Based on distance from the sensing object
  - Contact / No-contact

- Based on application industry
  - Medical / Manufacturing / Agriculture / etc.

- Based on measuring scale
  - Absolute / Relative

- Based on sensing mechanism
  - Thermoelectric / Electromechanical / Piezo resistive / Optic / Electric / Fluid mechanics / Photoelastic / etc.

- Based on sensing parameter
  - Position / Occupancy / Motion / Velocity / Force / Pressure / Flow / Humidity / Light / Temperature / Acoustic / Radiation / Chemical / Biosensors / etc.

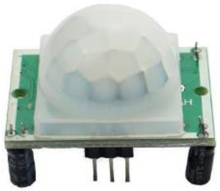
Sensor Types: What it measures

Sensor Type	Description	Example
Position	• Measures the position of an object • Position could be absolute/relative • Position sensor could be linear, angular, or multi-axis	• Proximity sensor • Potentiometer • Inclinometer
Occupancy	• Detects the presence of people and animals in a surveillance area • Generates signal even when a person is stationary	• Radar Sensor
Motion	• Detects the movement of people and objects	• Passive Infrared (PIR) Sensor

Ultrasonic Proximity Sensor

Infrared Proximity Sensor

Microwave Radar Sensor

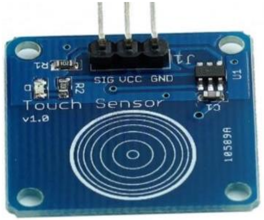


PIR Motion Sensor

Cont...

Sensor Type	Description	Example
Velocity and Acceleration	• Velocity sensor measures how fast an object moves • Acceleration sensor measures the changes in velocity	• Gyroscope • Accelerometer
Force	• Detects whether a physical force is applied and the magnitude of the force	• Tactile sensor • Viscometer
Pressure	• Measuring the force applied by liquids or gases • It is measured as force per unit area	• Barometer • Piezometer

Gyroscope



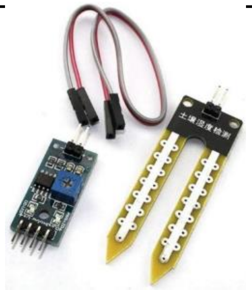
Capacitive Touch Sensor

Barometric Pressure Sensor

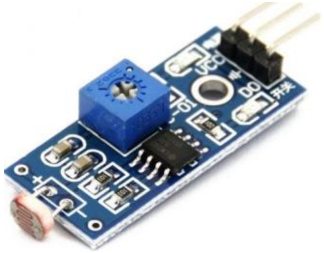
Cont…

Sensor Type	Description	Example
Flow	• Detects the rate of fluid flow through a system in given period of time	• Water meter • Anemometer
Humidity	• Detects amount of water vapour in the air • Can be measured in absolute/relative scale	• Hygrometer • Soil moisture sensor
Light	• Detects the presence of light	• LDR light sensor • Photodetector • Flame Sensor

Water meter



Soil moisture sensor



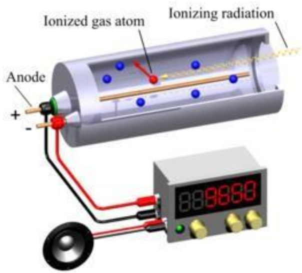
LDR light sensor

Flame sensor

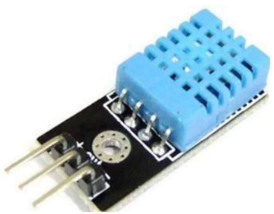
Cont…

Sensor Type	Description	Example
Radiation	• Detects the radiation in the environment	• Neutron detector • Geiger-Muller counter
Temperature	• Measures the amount of heat or cold present in the system • Two type: contact / non-contact	• Thermometer • Temperature gauge • Calorimeter

Neutron detector



Geiger-Muller counter

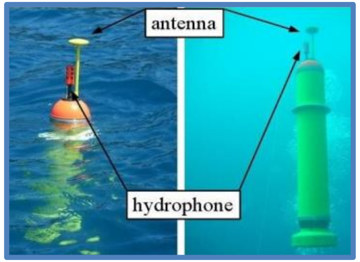


Temperature Sensor

Thermo-Hygrometer

Cont…

Sensor Type	Description	Example
Acoustic	• Measures sound level	• Microphone • Hydrophone
Chemical	• Measures the concentration of a chemical (e.g. CO_2) in a system	• Smoke detector • Breathalyzer
Biosensor	• Detects various biological elements, such as organisms, tissues, cells, enzymes, antibodies, nucleic acid, etc.	• Pulse oximeter • Electrocardiograph (ECG) • Blood glucose biosensor

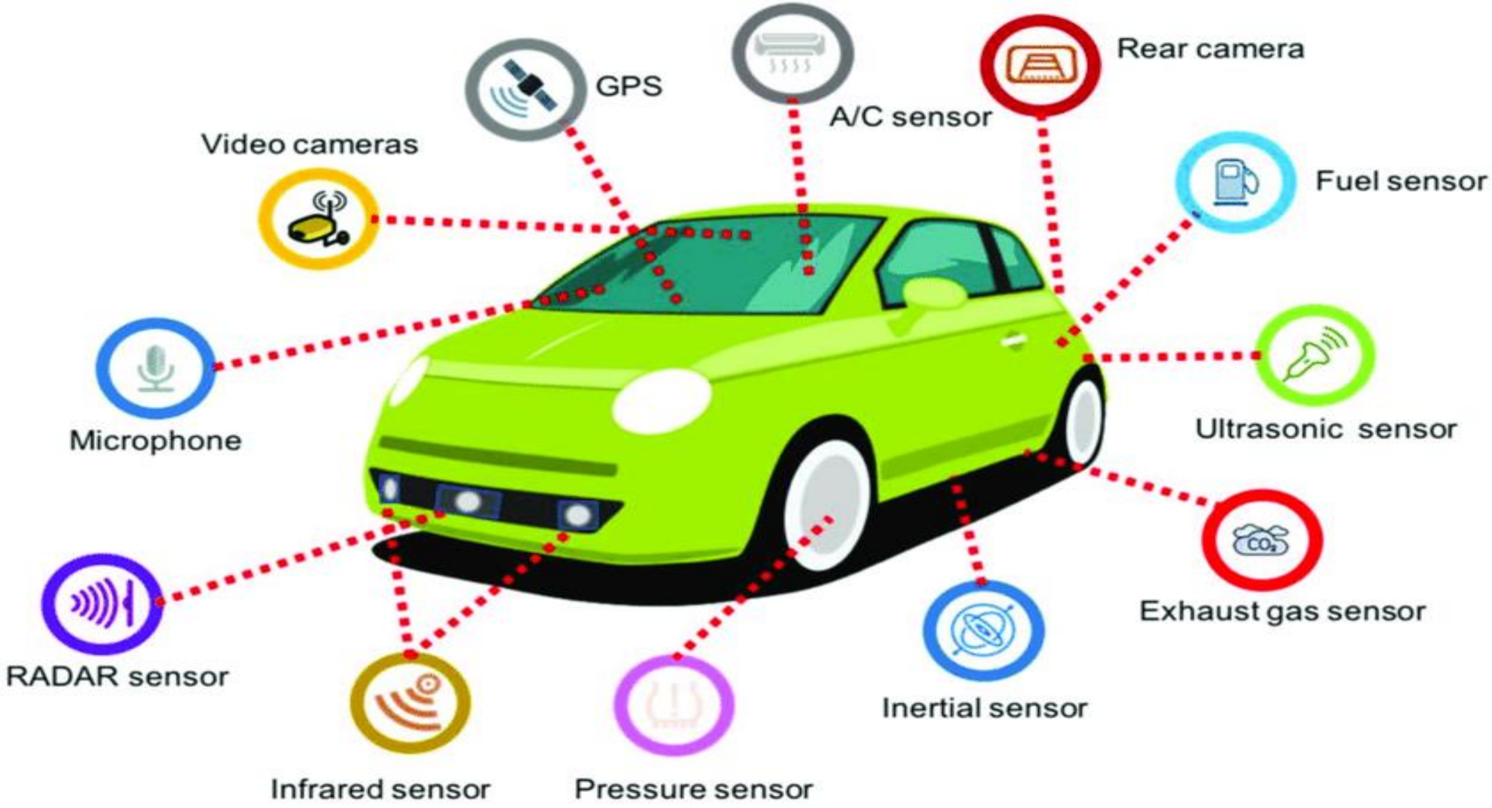


Hydrophone

Breathalyzer

Pulse oximeter

Sensors in a Smart Car

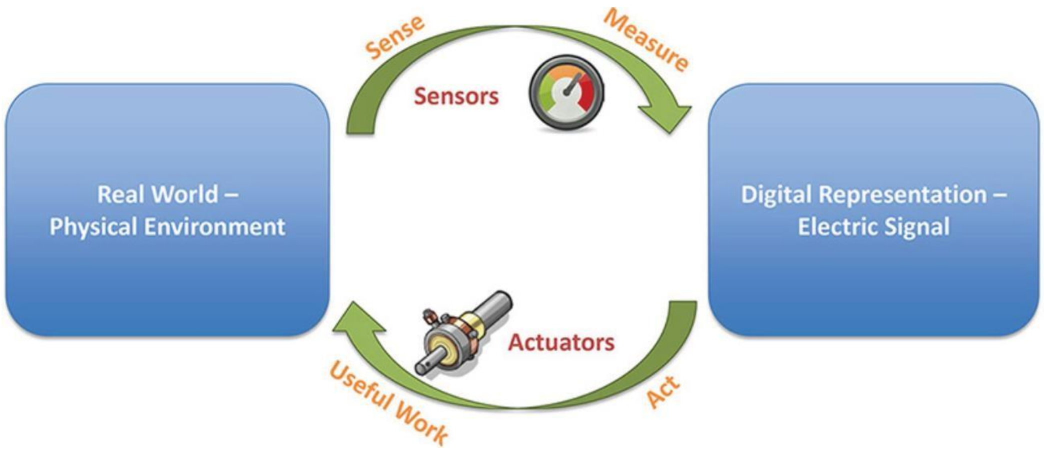


Sensors in a Smartphone

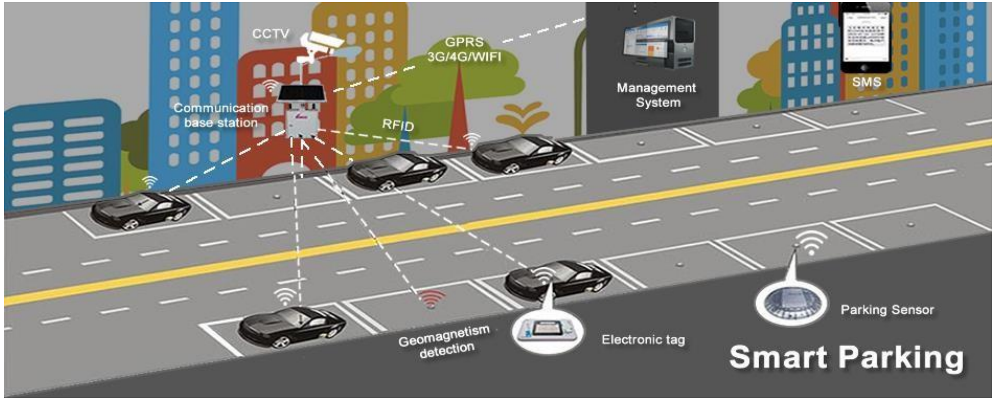
Actuators

- Sensors are designed to sense and measure the surrounding environment

- Actuators receive some type of control signal (commonly an electrical signal or digital command) that **triggers a physical effect**, usually some type of motion, force, and so on.

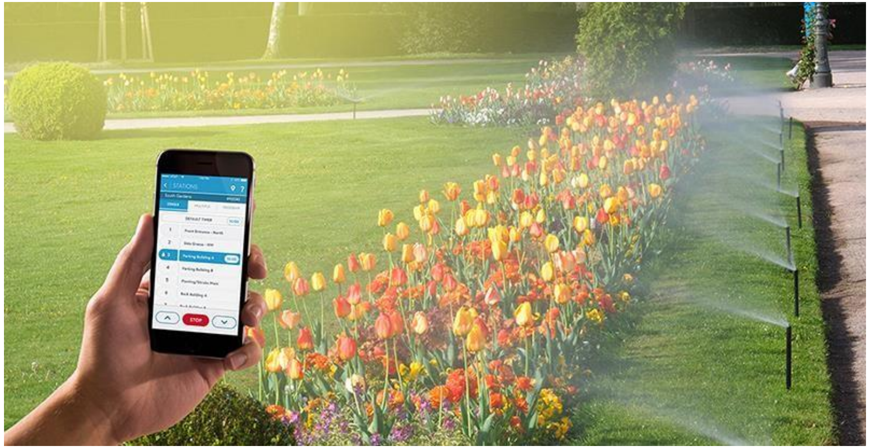


IoT based Automated Systems



Smart Parking System

Smart Irrigation System



Actuator Classification

- Common ways to classify actuators:

  ✓ Type of motion they produce
    - e.g. linear, rotary, one/two/three axes

  ✓ Power output
    - e.g. high power, low power, micro power

  ✓ Binary / Continuous output
    - Based on number of stable-state outputs

  ✓ Area of application
    - Specific industry or vertical where they are used

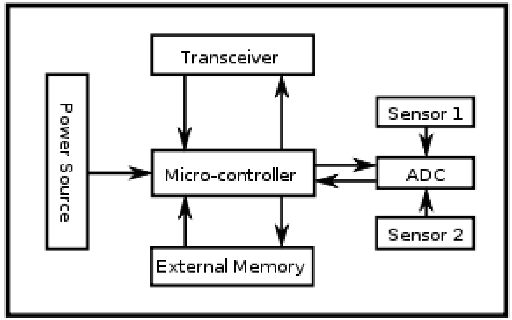
  ✓ Type of energy
    - e.g. mechanical energy, electrical energy, hydraulic energy, etc.

Actuators by Energy Type

Type	Examples
Mechanical actuators	Lever, Screw jack, Hand crank
Electrical actuators	Thyristor, Bipolar transistor, Diode
Electromechanical actuators	AC motor, DC motor, Step motor
Electromagnetic actuators	Electromagnet, Linear solenoid
Hydraulic and Pneumatic actuators	Hydraulic cylinder, Pneumatic cylinder, Piston, Pressure control valve, Air motor
Smart material actuator (includes thermal and magnetic actuators)	Magnetorestrictive material, Bimetallic strip, Piezoelectric bimorph

Smart Objects

- It is the building blocks of IoT
- Smart object has the following five characteristics:
  - Sensor(s) and/or Actuator(s)
  - Processing unit
    - For acquiring sensed data from sensors,
    - processing and analysing sensing data,
    - coordinating control signals to any actuators, and
    - controlling many functions (e.g. communication unit, power unit).
  - Memory
    - Mostly on-chip flash memory
    - user memory used for storing application related data
    - program memory used for programming the device
  - Communication unit
    - Responsible for connecting a smart object with other smart objects and the outside world (via the network using wireless/wired communication)
  - Power source
    - To powered all components of the smart object



TelosB Mote

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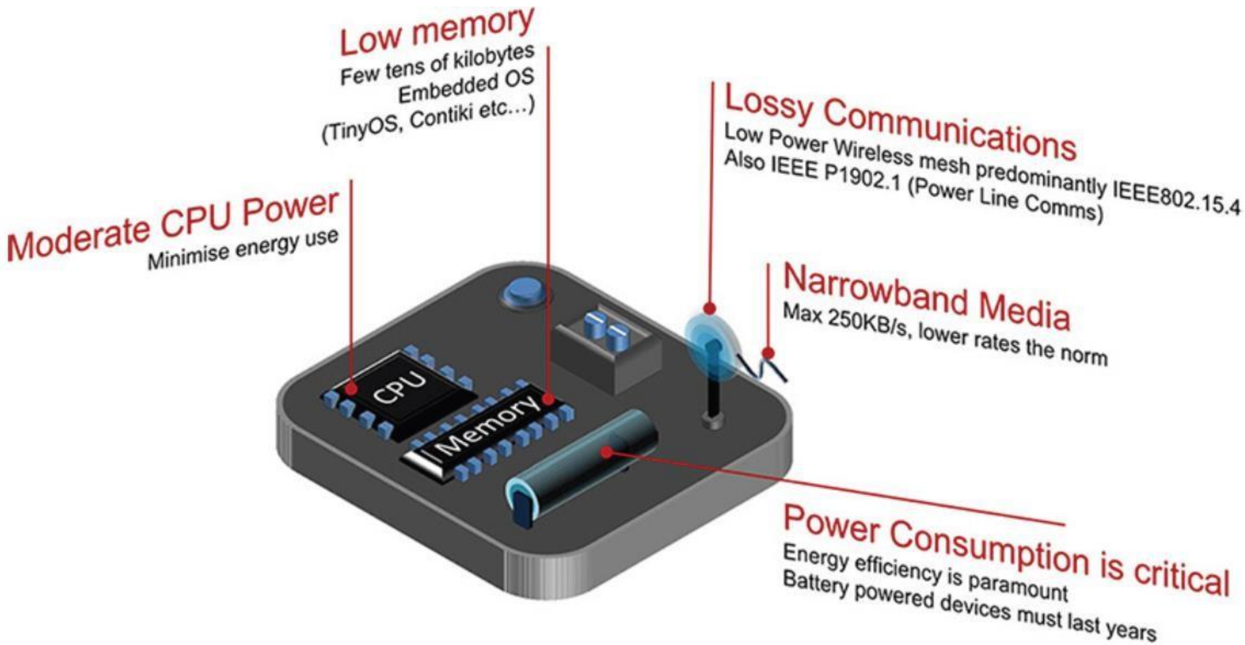
Low memory
Few tens of kilobytes
Embedded OS
(TinyOS, Contiki etc…)
Moderate CPU Power
Minimise energy use
Lossy Communications
Low Power Wireless mesh predominantly IEEE802.15.4
Also IEEE P1902.1 (Power Line Comms)
Narrowband Media
Max 250KB/s, lower rates the norm
Power Consumption is critical
Energy efficiency is paramount
Battery powered devices must last years
CPU
Memory

Present Trends in Smart Objects

- Size is decreasing
- Power consumption is decreasing
- Processing power is increasing
- Communication capabilities are improving
- Communication is being increasingly standardized


Question
&
Answer